Project 3: Milestone 3 (Final White Paper)

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Topic

Predicting Census Income Categories and Interpreting Socioeconomic Drivers (Adult Census Income Dataset).

Abstract

This white paper reports an end-to-end modeling study using the Adult Census Income dataset to predict whether an individual's annual income exceeds $50,000. After standardizing the schema, mapping literal "?" to an explicit Unknown category, and creating a stratified 80/20 train/test split, I trained two baseline models inside leak-safe pipelines: Logistic Regression and Random Forest. On the held-out test set, Logistic Regression achieved accuracy 0.854, precision 0.739, recall 0.610, F1 0.668, and ROC-AUC 0.904. Random Forest performed similarly on accuracy and AUC with a modest recall uplift (accuracy 0.853, precision 0.728, recall 0.623, F1 0.672, ROC-AUC 0.900). Interpretation via LR coefficients and RF importances highlights capital gain, marital status, education, hours worked, and age as key drivers. Subgroup evaluation shows small but meaningful performance differences (for example, F1 of 0.67 for males vs 0.64 for females), underscoring the need for calibrated thresholds and fairness-aware review before operational use. Figures referenced herein include EDA snapshots, ROC curves, confusion matrices, and model interpretability plots, supporting a transparent and reproducible analysis.

Business Problem

The goal of this project is to develop and evaluate interpretable models for classifying whether an adult's annual income exceeds $50,000, based on routine census metrics. The objective is twofold: achieve reliable predictive performance on held-out data, and to provide transparent explanations of the key drivers associated with higher income. These insights can inform workforce development agencies, policy makers, and employers seeking to design interventions that promote economic opportunity and growth. Success is measured with standard classification metrics, including accuracy, precision, recall, F1, and ROC-AUC, combined with clear model interpretation and a discussion of subgroup performance.

Background/History

The Adult Census Income dataset originates from the 1994 U.S. Census Bureau Current Population Survey. It has become a widely used benchmark in tabular classification. The target is a binary indicator of income category (>50K vs <=50K), derived from self-reported earnings. The dataset includes a mix of demographic and employment variables (for example: age, education, marital status, occupation, and hours working per week). It also exhibits some noticeable class imbalance.

Data Exploration

I began by loading the dataset and standardized its schema for further processing. Categorical fields contained a literal “?” to denote missingness indicating a lack of response. I treated those missing values by mapping explicitly to “Unknown”. I normalized column names to snake case. The binary target “income_binary” encodes “>50K” as 1 and “<=50K” as 0. The cleaned table includes 32,561 rows and 16 columns, spanning both numeric features (“age”, “hours_per_week”, “capital_gain”, “capital_loss”) and categoricals (“workclass”, “education”, “marital_status”, “occupation”, “relationship”, “sex”, “race”, and “native_country”). The class distribution was imbalanced, with roughly 24% of observations in the positive class. I made a stratified train/test split (26,048 / 6,513 rows) in order to preserve that ratio across the test and training splits. A high-level EDA confirmed the heavier right tails in age among higher-income observations, and a higher central tendency and spread for weekly hours in the >50K group.

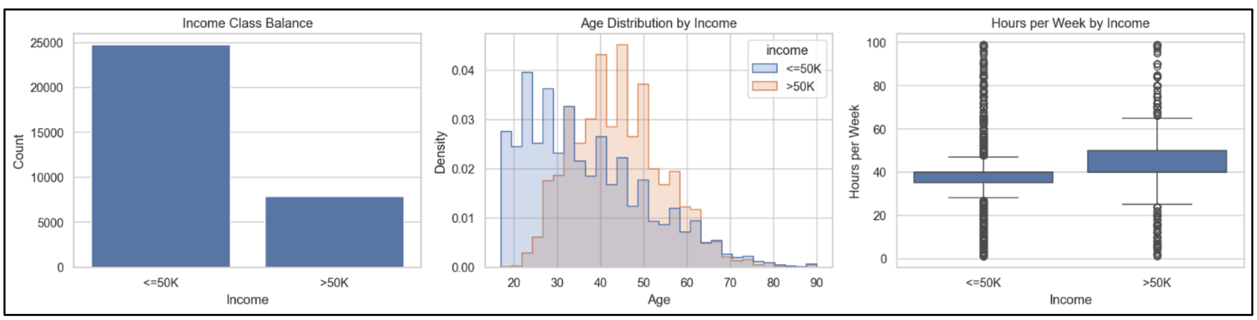

Figure 1: EDA overview showing class balance and representative distributions for age and “hours_per_week” by income.

Methods

Data Preparation

I standardized all column names to snake case, trimmed whitespace in categorical fields, and mapped literal "?" responses to an explicit Unknown category to preserve the information content of non-response. The binary target "income_binary" encodes ">50K" as 1 and "<=50K" as 0. I created a stratified 80/20 train/test split on the binary target to maintain class balance across splits and dropped the original string label during training to prevent leakage.

Modeling

I trained two baseline models with reproducible pipelines. Logistic Regression used L2 regularization (max_iter=1000 and a fixed random seed) with standardized numeric features and one-hot encoded categoricals. Unknowns were ignored at inference. Random Forest used 200 trees with default splitting parameters, and one-hot encoded categoricals into a dense matrix. Both pipelines produced 107 engineered features after encoding, verified on the training split.

Evaluation and Interpretation

Evaluation was focused on held-out test performance using accuracy, precision, recall, F1, and ROC-AUC. These metrics were evaluated alongside confusion matrices and ROC curves. I complemented aggregate metrics with subgroup evaluation by "sex" and "race" to surface any performance disparities. For interpretability, I extracted and visualized the largest-magnitude Logistic Regression coefficients and the top Random Forest feature importances to identify influential drivers of model predictions.

Analysis

On the test set, Logistic Regression (LR) achieved accuracy 0.854, precision 0.739, recall 0.610, F1 0.668, and ROC-AUC 0.904. The confusion matrix indicates more false negatives than false positives at the default threshold, reflecting class imbalance and conservative decisioning for the positive class. Random Forest (RF) performed similarly overall with accuracy 0.853, precision 0.728, recall 0.623, F1 0.672, and ROC-AUC 0.900. RF modestly improves recall relative to LR while maintaining comparable accuracy and AUC, a common pattern for tree ensembles on mixed-type tabular data.

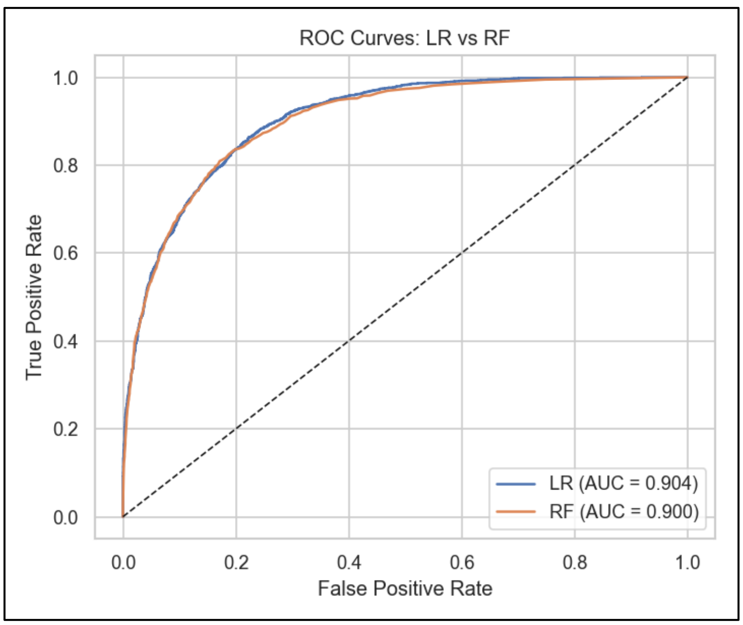

Figure 2: ROC curves for LR and RF. Both models achieve strong separability.

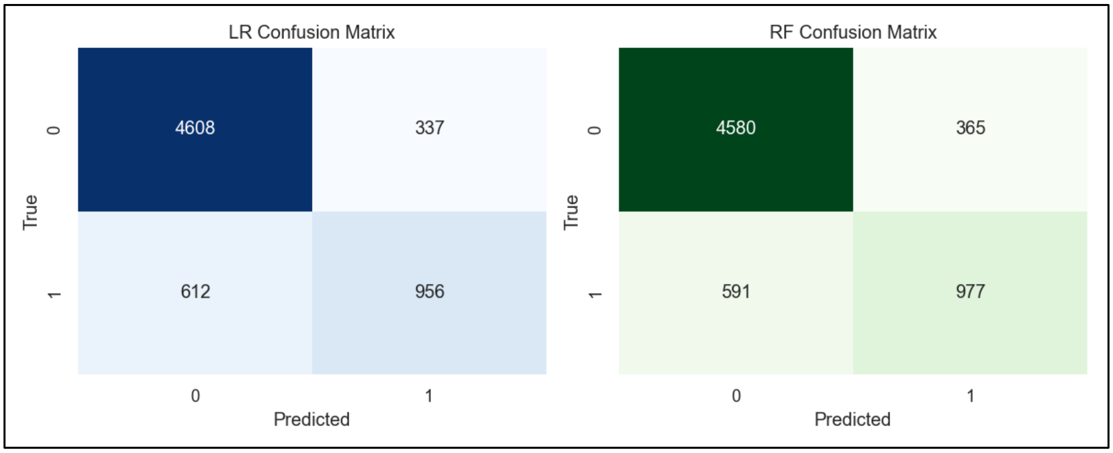

Figure 3: Confusion matrices show asymmetric errors with more false negatives. RF reduces missed positives modestly.

Interpretability aligns with domain expectations. The LR coefficient plot highlights a small set of high-magnitude signals (certain education levels, marital status categories, weekly hours, capital gains, and select occupations) that strongly increase or decrease the log-odds of earning >50K. RF importances, while less directionally explicit, corroborate that a handful of variables exert outsized influence among the engineered features, including hours worked, capital gains, age, and education-related attributes. Together, these views provide both directionality (LR) and non-linear influence (RF), enabling richer stakeholder discussions about drivers.

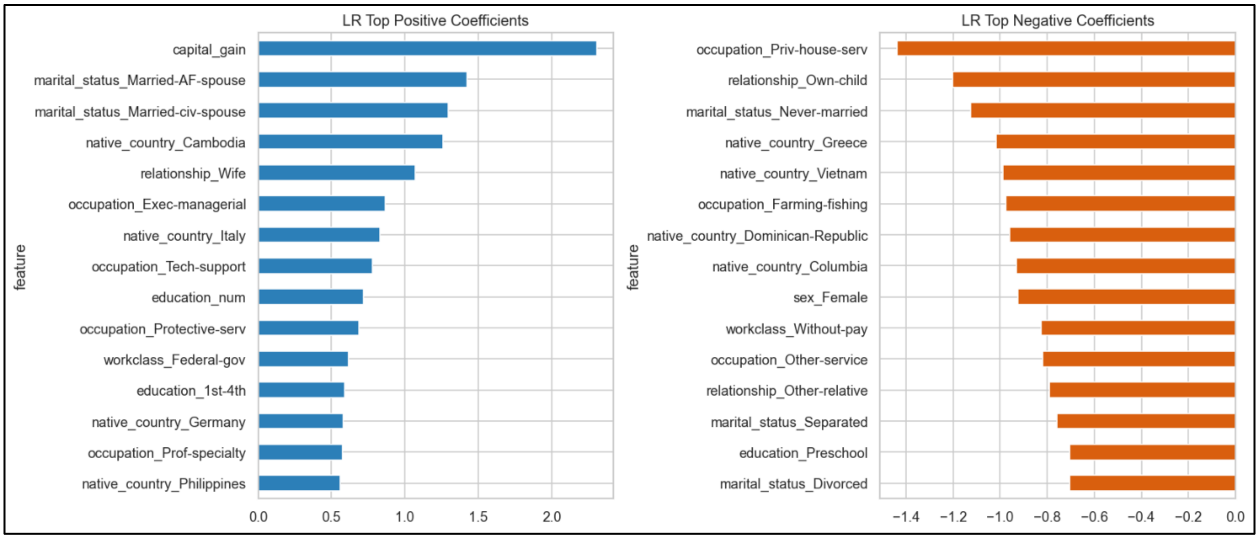

Figure 4: Largest-magnitude LR coefficients reveal features that most increase/decrease odds of >50K.

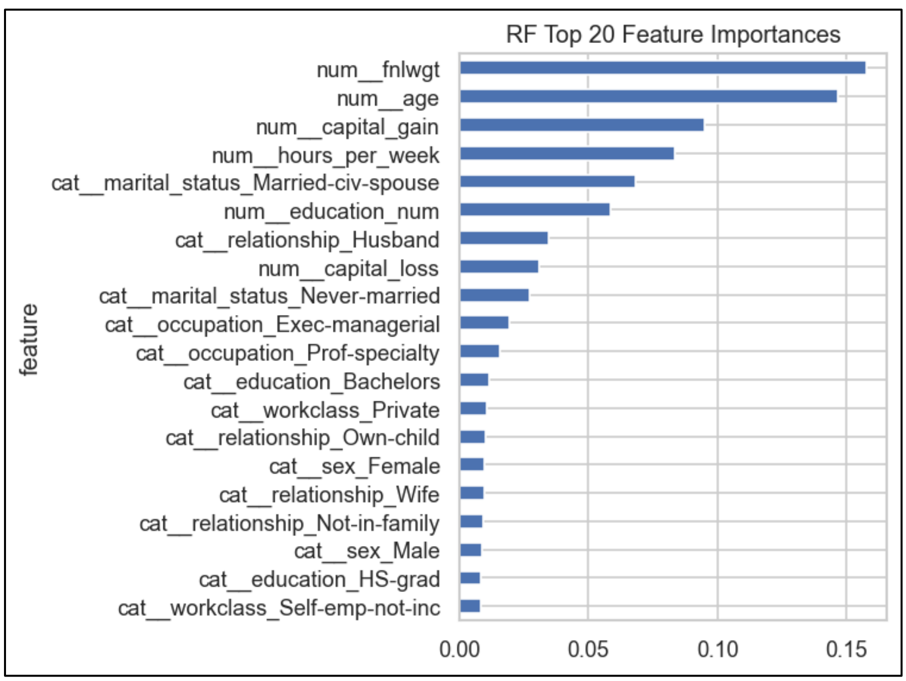

Figure 5: RF importances indicate high-influence variables among the 107 engineered features.

I also examined subgroup performance by sex and race. Both models showed higher F1 for males than females (approximately 0.67 vs 0.64). RF improved recall for several subgroups relative to LR, including a noticeable uplift for the Black subgroup (F1 ≈ 0.65 vs ≈ 0.60 under LR), while maintaining overall AUC and accuracy. Smaller subgroups exhibited greater variance due to limited support. These trade-offs underscore the need to consider cost-sensitive thresholds and fairness checks before any operational use.

Interpretation of Results

Taken together, the models provide strong probability ranking (AUC of approximately 0.90) and competitive classification performance. They exhibit modest asymmetry in errors, favoring lower false positives over missed positives at the default 0.5 threshold. Random Forest's recall was slightly higher than that of Linear Regression, though it trades a small amount of precision. The dominant drivers (capital gains, education attainment, hours worked, marital status, age, and specific occupations) align with socioeconomic intuition. Since subgroup F1 varies across sex and race, decision thresholds should be calibrated and stress-tested for equity impacts prior to deployment. A single threshold may not optimally balance errors for all cohorts. In policy or outreach contexts, these results are best used to inform population-level strategies rather than to make deterministic decisions about individuals.

Conclusion

The cleaned and standardized Adult dataset supports transparent modeling of income categories with strong baseline performance. Logistic Regression and Random Forest produce similar accuracy and AUC, with RF offering slightly higher recall. Interpretation via LR coefficients and RF importances converges on intuitive drivers like education, hours worked, capital gains, and age. Subgroup analysis highlights modest but meaningful differences across sex and race, suggesting that future work should incorporate calibrated thresholds, fairness-aware evaluation, and possibly constrained optimization to manage performance trade-offs responsibly.

Assumptions

This analysis assumes that the Adult dataset is sufficiently representative of the targeted populations for aggregate insights, that reported income categories and covariates are accurate, and that the relationship between features and the outcome is stable over time. The train/test split is assumed independent and identically distributed without temporal or household dependencies, and the preprocessing correctly handles all missingness via explicit “Unknown” categories rather than imputation that could bias distributions.

Limitations

The data are a historical snapshot and may not reflect current labor markets. Sensitive attributes and their proxies can encode historical bias, complicating causal interpretations. Class imbalance leads to asymmetric errors at the default threshold. Certain variables (e.g., capital gains/losses) are highly skewed with many zeros, which can challenge some models. This draft does not include calibration analysis, cost-sensitive thresholding, or comprehensive fairness diagnostics beyond subgroup summaries. Finally, model explainers here are coefficient magnitudes and impurity-based importances; they do not in themselves establish causality.

Challenges

The main challenges in this project involve handling class imbalance, standardizing mixed feature types, and preventing data leakage while still capturing information from non-response. I also had to engineer pipelines that keep numeric features properly scaled for LR while producing dense encodings that work well for RF, which turned out to matter for both correctness and efficiency. Choosing a decision threshold that balances false negatives and false positives also required careful consideration/tuning.

Future Uses/Additional Applications

Aggregate insights from these models can inform policy simulations (including: how increased educational attainment or hours worked correlate with higher earnings categories), workforce program targeting, and exploratory research into socioeconomic mobility. With appropriate safeguards, the models can support triage workflows for outreach or counseling, provided that predictions are never used deterministically on the individual level.

Recommendations

In the near term, maintain both LR and RF baselines: use LR when directional interpretability is paramount and RF when modest recall improvements justify slightly reduced transparency. Introduce probability calibration and cost-sensitive thresholding to align decisions with stakeholder preferences (for example, prioritizing recall in outreach contexts). Expand subgroup evaluation with formal fairness metrics and confidence intervals. Document model cards detailing intended use, limitations, and monitoring plans. Prefer population-level storytelling over individual predictions in any stakeholder communication.

Implementation Plan

I keep the pipeline fully reproducible by employing schema checks, fitting all preprocessing components on the training split only, and training LR and RF baselines with fixed seeds and tracked hyperparameters. Evaluation runs on a held-out set with calibrated decision thresholds, and we log metrics, confusion matrices, and subgroup performance reports for review. I then package the entire workflow for batch scoring behind an access-controlled interface. Before deploying anything, I recommend running a fairness review, adding monitoring for data drift and performance degradation, and setting up a scheduled retraining plan with appropriate governance.

Ethical Assessment

Income prediction involves sensitive data points, such as sex, race, and native country. Sometimes, these kind of data points may reflect inequities. Therefore, my analysis will really emphasize transparency, with disclaimers to avoid using the model for individual-level decisions, and I will try to highlight any detected disparities. Sensitive features will be examined both with and without inclusion to understand their influence, and results will be contextualized to discourage unfair or stigmatizing interpretations. The dataset is publicly available and de-identified, mitigating privacy concerns. Documentation will clearly state limitations and ethical usage considerations for stakeholders.

Appendix (Supporting Documentation)

- A supplementary document with anticipated questions and answers from the audience is attached, entitled *DSC_680_Koyrakh_Project3_QA.pdf*.
- The full project code is provided with the submission as a Jupyter Notebook entitled *DSC680_Koyrakh_Project3_Notebook.ipynb*.
- A presentation of this project for stakeholders is attached:
    - *DSC680_Koyrakh_Project_3_Presentation.mp4*
    - *DSC680_Koyrakh_Project_3_Slideshow.pptx*

References

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